APPENDIX B

Answers to Exercises

Chapter 2: (1) π-mesons, conversion of excess energy into mass. **(2a)** gravitation, the electromagnetic force, the weak interaction force, and the strong interaction force. (**2b**) See e.g. Tables 2.2 and 2.3. **(3a)** Bosons are a group of 'particles' to which the photon and mesons belong. The bosons are the carriers of forces. When two fermions interact they continually emit and absorb bosons. The bosons have an even spin (0, 1, etc.), they do not obey the Pauli principle, and they do not require the formation of antiparticles in their reactions. Particles of nonintegral spin are called fermions because they obey the statistical rules devised by Fermi and Dirac, which state that two such particles cannot exist in the same closed system (nucleus or electron shell) having all quantum numbers the same (referred to as the Pauli principle). Fermions are the building blocks of nature. Fermions can be created and destroyed only in conjunction with an antiparticle of the same class. For example if an electron is emitted in β-decay it must be accompanied by the creation of an antineutrino. Conversely, if a positron − which is an antielectron − is emitted in the β-decay, it is accompanied by the creation of a neutrino. (**3b**) Answer is included in, and follows from, answer to (a). **(4)** Radiation pressure, recoil in photon and γ-emission, energy increase and decrease when falling into or moving out against a gravity field. **(5)** Several possibilities, e.g.: Reaction with ^{1}H to produce a fast neutron and a positron. Then detection of annihilation γ and neutron, $^{37}Cl + n = {}^{37}Ar + e^{-}$ followed by collection of Ar and detection of its decay, $^{71}Ga + n = {}^{71}Ge + e^{-}$, conversion to GeH_4, collection and detection of its decay.

Chapter 3: (1) 1.553×10^{22}. **(2)** 0.100. **(3a)** 2111 m s^{-1}. **(3b)** 0.0462 V **(4)** 472 V **(5)** 11.00934 u. **(6)** 0.129 nm **(7a)** 3.032×10^{17}. (**7b**) 3322 K **(8)** 2.080. **(9)** 4. (**10**) 0.030 m (**11a)** By series expansion $e^{\delta} = 1 + \delta + \delta^2/2 + \ldots$but $\alpha = 1 + \delta$ hence $e^{\delta} \simeq \alpha$. (**11b)** 1.097. (**11c)** 22. (**12)** 5.742 SWU.

Chapter 4: (1) 8.261 MeV **(2)** 1.581×10^{6}. (**3**) 47371 km (**4**) Fusion gives 7 times more per gram. (**5**) −1.02 MeV, −2.76 MeV, and −1.04 MeV (**6**) 6.262 MeV u^{-1}, 8.366 MeV u^{-1}, 8.734 MeV u^{-1}, and 7.489 MeV u^{-1}. (**7**) 4.8, 12.6, 26.4, and 91.8.

Chapter 5: (1) 0.088 MeV, ^{235}U. **(2a)** 0.87 MeV **(2b)** 0.46 MeV (measured 0.51 MeV). **(3)** 99 eV **(4)** 1.26 keV **(5)** 547 keV (measured 662 keV). **(6)**. 1.28×10^{9} years **(7)** 5.9×10^{9} years **(8)** 2.21×10^{22} J $year^{-1}$. **(9a)** 16.9 GBq. **(9b)** 3.0×10^{-10} m^3 **(10a)** Multiply (5.25) by $\mathbf{c}^2$. **(10b)** $m_{\gamma}^{0} = 0 \rightarrow m_{\gamma} = E_{\gamma}/\mathbf{c}^2$ hence $E_{\gamma}(E_{\gamma}/\mathbf{c}^2) = E_{d}m_{d}$. **(11a)** 1.46×10^{4} Bq. **(11b)** 8.09×10^{8} atoms. **(12)** $t_{½} = 104$ h and 99 Bq, $t_{½}$ 4.4 h and 1030 Bq. **(13)** R_0 507 cpm for $t_{½}$ 107 min, R_0 440 cpm for $t_{½}$ 24 min (cf. ^{298}At $t_{½}$ 98 min, ^{206}At $t_{½}$ 31 min) **(14)** 2.3×10^{5} years.

Chapter 6: (1a) 10. **(1b)** 10. **(2a)** 2.19×10^6 m s^{-1}. **(2b)** 0.003%; no. **(3)** 21 MHz **(4)** $2.689 \times 2/3$. **(5)** 1i, 2g, 3d, 4s; 56 nucleons. **(6)** 283 MeV, **(7a)** 7/2, $3.81 \times (1/5)$. **(7b)** 5, $4.50 \times (2/5)$. **(7c)** 5/2. **(8a)** 0.108. **(8b)** 7/2 (also observed). **(9)** 1p½. **(10a)** 43.8 and 307 keV **(10b)** 3.47 fm (8.68 fm from (4.14), meas. 5.48 fm). **(11)** 0.460 T **(12)** 3.2×10^9 years **(13)** 8.86×10^{10} years, hindrance factor 1.2.

Chapter 7: (1) From (7.12) R_{Al} 5.8 mg cm^{-2}, and from Fig. 7.6 ∼6 mg cm^{-2}, diff. ∼3%. **(2)** E_α 2.0 MeV **(3)** 0.71 mm **(4)** e^- 3.8 m, H^+ 23 cm, α 3.3 mm **(5)** Al 0.030, Ni 0.012, Pt 0.0074 mm **(6)** 6.54×10^7 γ m^{-2} s^{-1}. **(7)** ^{3}H 5 μm, ^{14}C 270 μm, ^{32}P 5.3 mm, ^{90}Sr 1.1 mm **(8)** 2.92×10^8 m s^{-1}. **(9)** Lead $105 000, concrete $3730. **(10)** Na_K 57.53, Na_L 58.54, I_K 25.43, I_L 53.80 keV **(11)** 72% and 52%. **(12)** 28, 39, and 49 g cm^{-2}. **(13a)** Emax 0.62 MeV **(13b)** ∼0.8 MeV **(14a)** 9.7 days. **(14b)** 19 min **(15)** 4.9 m **(16a)** 5.5 cm **(16b)** 11.9. **(16c)** 2.5.

Chapter 8: (1a) $\mathbf{1.47 \times 10^5}$. **(1b)** 2.94×10^4. **(1c)** 972. **(2)** ∼1.2%. **(3)** 3.8 pGy, **(4)** 0.34%. **(5)** ∼11%. **(6)** ^{90}Sr 0.10, ^{90}Y 0.068, ^{3}H 2.5 keV μm^{-1} **(7)** 6.3 mGy s^{-1} **(8)** 38 pF.

Chapter 9: (1a) 1.33 GBq. **(1b)** 2.5%. **(2a)** 25 μm **(2b)** 1.05×10^5 μm^{-1} **(2c)** 4.6 nm **(2d)** 3.8 eV atom^{-1}. **(2e)** 29 400 K **(3)** 0.11 μM **(4)** 0.88 Bq. **(5)** 5.8 V **(6a)** 509 μs. **(6b)** 12.53. **(6c)** 17.84. **(7)** 3.2×10^{10} counts. **(8)** 45 ns. **(9a)** 6.6 μg cm^{-2} **(9b)** 1.59. **(9c)** 71%. **(10a)** 9.33 ± 0.11 cps. **(10b)** ±0.072 cps. **(11)** 0.5%.

Chapter 10: (1a) 37.3 MeV **(1b)** 1305 m s^{-1}, 0.035 eV **(2)** 47.4 fm **(3a)** 32.9 b sr^{-1}. **(3b)** 0.423 m **(4)** 0.75 MeV **(5a)** 39.5 fm **(5b)** 7.56 fm **(6a)** 8.03 MeV **(6b)** 1.64 MeV **(7)** 3.09×10^6 m s^{-1}. **(8)** 3.57×10^{28} kg **(9)** 77.3 MeV **(10)** Q (0.87 MeV) > 0, He thus unstable. **(11a)** −5.85 MeV **(11b)** 5.85 MeV **(11c)** 2.96 MeV **(11d)** 8.81 MeV.

Chapter 11: (1) 9.9×10^{-8} s^{-1}. **(2)** 6.0%. **(3a)** 3.17×10^{14}. **(3b)** 1 in 19,900. **(3c)** max 29 kW **(4)** 36.7 m^{-1} **(5)** yes. **(6)** light 103.4 MeV, heavy 74.3 MeV **(7)** 1.78 MeV **(8)** $E_\gamma \geq 11.46$ MeV **(9)** 2.7×10^8 K.

Chapter 12: (1a) 1.9 keV **(1b)** ∼1.8×10^{-30}. **(2a)** 12.3 × Lawson crit. **(2b)** 49 MW (3) 283 l **(4)** 3.35×10^{10} kg s^{-1}. **(5)** 1.68%. **(6)** Water power ∼9.8×10^7 W, fusion ∼1.2×10^{15} W.

Chapter 13: (1) Isotropic flux above 1 GeV. No variation with solar activity. **(2a)** High energy particles; ∼70% p, ∼20%, ∼0.7% Li + Be + B ions, ∼1.7% C + N + O ions, rest $Z > 10$. **(2b)** Mostly not. **(3)** 200−300 cps. **(4)** See Chapter 2, energy to mass conversion. **(5)** 0.131 Bq. **(6)** 9.76×10^6 years **(7)** average 266 Bq kg^{-1}. **(8)** 1.65×10^9 years **(9)** 5.0×10^8 years **(10)** 3.70×10^9 years **(11)** 1.17×10^{-3}, 1.41×10^{-3}, and 2.33×10^{-4} W m^{-2}. **(12a)** 193 t year^{-1}. **(12b)** 4.6×10^5 t year^{-1}. **(13)** ∼18, ∼53, and ∼211 years.

Chapter 14: (1) n, on ^{196}Hg, or n,2n on ^{198}Hg. **(2)** 93%. **(3)** ^{239}Pu from d,n in ^{238}U. **(4)** ^{248}Cm and FP; ^{253}Es and ^{249}Cm; ^{250}Cm and FP. **(5)** 2.30 n per fiss. **(6)** possibly ^{259}Sg. **(7)** Am(III) has larger ionic radius than Eu(III) and is thus less strongly solvated by water. **(8)** 0.085 W **(9)** 5 f^{14} 7 s^2. **(10)** Pu(III)/Pu(IV) ∼970 mV, Pu(IV)/Pu(V) ∼1050 mV, Pu(V)/Pu(VI) ∼930 mV; correct values in Fig. 14.7.

Chapter 15: (1) ^{3}H 163.5 Bq, ^{14}C 2.85 kBq, ^{40}K 4.40 kBq, Ra and daughters 0.93 Bq, sum 7.41 kBq/70 kg **(2)** 4.2×10^{-7}%. **(3)** 1.8×10^{-6}%. **(4a)** 124 μSv year^{-1}. **(4b)** 39 μSv year^{-1}. **(5)** 21 mSv. **(6a)** 4.37 MeV **(6b)** 698 million. **(7)** 1.14 μs.

Chapter 16: (1a) 0.329 cm **(1b)** 9.0 MeV **(1c)** ~^{38}Ar. **(2)** Between 1st and 2nd. **(3a)** 10 kW **(3b)** 228 m s^{-1}. **(3c)** 2646 m s^{-1}. **(4)** 7.6×10^{11} n m^{-2} s^{-1} **(5)** 4.105 m **(6a)** $L_2 = 1.41$ cm, $L_3 = 1.73$ cm **(6b)** 111.4 MHz **(7)** H^+ 11.8 MeV, 0.80 T; D^+ 23.6 MeV, 1.598 T: He^{2+} 46.9 MeV, 1.588 T **(8)** 1.01×10^{10} s^{-1}. **(9)** 8.53×10^3 g **(10a)** 35. **(10b)** 78.8 MeV **(10c)** 124 MeV.

Chapter 17: (1) 28 TBq g^{-1} Na. **(2)** 10.6%. **(3)** 0.85 TBq. **(4a)** 1.56×10^7 s **(4b)** 0.145. **(5)** 4873 cps. **(6)** 10.5 d.

Chapter 18: (1) 4.7 l **(2)** 24%. **(3)** 21.0%. **(4)** 1.1 mg **(5)** 45 ppm Ga. **(6)** 20.4%. **(7)** 3.32×10^{-4} s^{-1} M^{-1} **(8)** Calculated $\log(\beta_n)$ values are: 6.1, 10.6, 13.8. **(9)** $\log \beta_4 = 15 \pm 1$ and $\log K_{DC} = 0 \pm 0.3$. **(10)** 29 m^3 **(11)** −0.24 V **(12)** $D = 1$ for V at pH = 0.5 and for U at pH = 2.6; optimal separation at pH = 1.55. **(13)** 99.9% U with 9.8% La.

Chapter 19: (1) $465 million. **(2)** 0.765 kg ^{235}U. **(3a)** 7 oil tankers (7.1). **(3b)** One train car (8.9 t). **(3c)** 4 train cars (75.4 t). **(4)** 21 collisions (20.63). **(5)** 1.898. **(6)** $\eta = 2.022, f = 0.750, k_\infty = 1.451$. **(7)** 1.257 m **(8)** 2.131 m side. **(9a)** $\Lambda = 0.607$. **(9b)** $r = 0.256$ m **(10)** 23.1%. **(11)** ~1100 times. **(12)** 3.71×10^{12} atoms s^{-1}. **(13)** 9.0 h **(14)** 3.07 kg **(15)** 49 MW (16) 8.4×10^{17} fusions or ~1.7% consumed of all D^+.

Chapter 20: (1) ~34000 kg **(2)** 14600 m^3 **(3a)** 0.64 s **(3b)** 193 GJ **(4)** ^{36}Cl, ^{37}Cl, ^{40}K, and ^{42}K. **(5a)** ^{232}Th(n,2n)^{231}Th(β$^-$)^{231}Pa(n,γ)^{232}Pa(β$^-$)^{232}U. **(5b)** Yes. **(6)** 60%. **(7a)** Rest heat cooling to the sea. **(7b)** HV-stability more important. **(8)** Recirculated back to off-gas system (multiplies decay time).

Chapter 21: (1) 674 and 1200 °C. **(2a)** 1.40 kg **(2b)** Effective cross-sections (Fig. 19.5), self-screening, and capture to ^{240}Pu. **(3)** 152 d **(4a)** 2.333 kg ^{235}U. **(4b)** 1.141 kg ^{235}U. **(5)** See Fig. 19.17. **(6a)** 31.6 kg FP. **(6b)** 0.91%. **(7a)** ~202 TBq. **(7b)** Ce, Pr, Nb, Zr, Y, Sr, and Ru. **(8a)** 8.7×10^5. **(8b)** 3.9×10^2. **(9)** 404 MBq ^{238}Pu, 94 MBq ^{240}Pu. **(10)** 9.04×10^{15}. **(11a)** 95 kW **(11b)** 34 cylinders. **(11c)** 1.5 reactor years. **(12)** Ru 710 kg, 10.2 TBq kg^{-1}; Rh 153 kg, 47 TBq kg^{-1}; Pd 456 kg, 2.92 MBq kg^{-1}.

Chapter 22: (1) See Section 21.13.1. **(2)** None. **(3)** NpO, Pu^{4+}. **(4a)** See Section 21.13.3. **(4b)** ASSE salt dome, WIPP bedded salt. **(5)** $10^{13.8}$. **(6)** 3.7×10^5. **(7a)** $E = 0.739$ 0.236 pH. **(7b)** $Eh > 0.677$. **(8a)** 2.92 kBq. **(8b)** No.

Appendix A: **(1a)** 1.5%. **(1b)** 99.9%. **(2)** Use smaller solution for P; $\theta_{extr} = 8.60$, $\theta_{wash} = 94.6$.